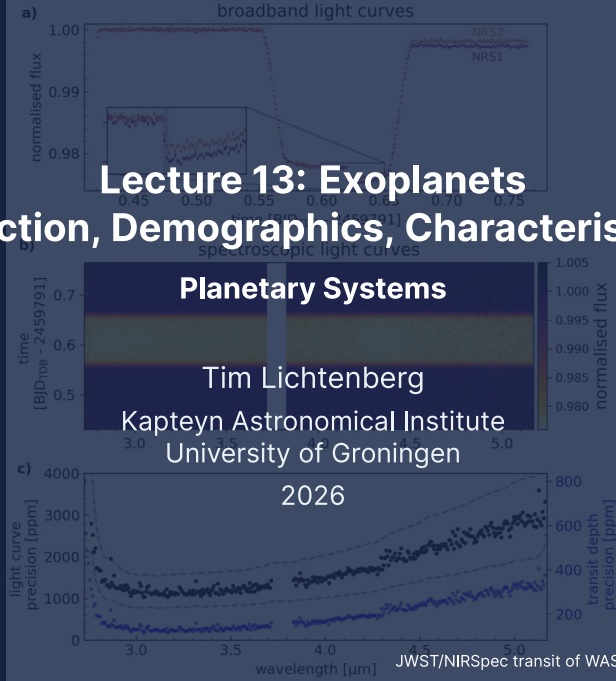


Lecture 13: Exoplanets

Detection, Demographics, Characterisation



Learning Objectives

By the end of this lecture, you will be able to:

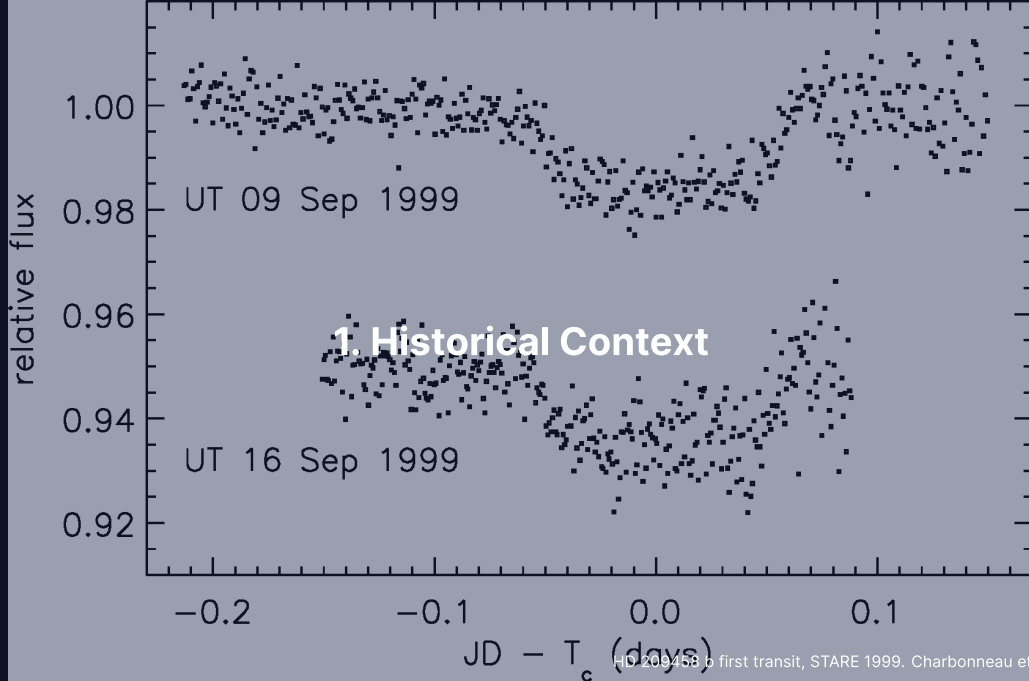
- Describe the main exoplanet detection methods and their biases
- Combine transit and radial-velocity geometry into a bulk density, and read the radius valley
- Evaluate JWST-era atmospheric results and biosignature claims

From zero confirmed exoplanets in 1991 to more than 6000 in 2026.

Recap: Where We Are in the Course

- Lectures 9–11 surveyed the geology, atmospheres, and interiors of the solar planets
- Lecture 12 decoded small bodies and meteorites as planet formation fossils
- This lecture turns outward to thousands of exoplanets to test if these patterns are universal

Today: How do we detect, weigh, and characterise worlds beyond our solar system?

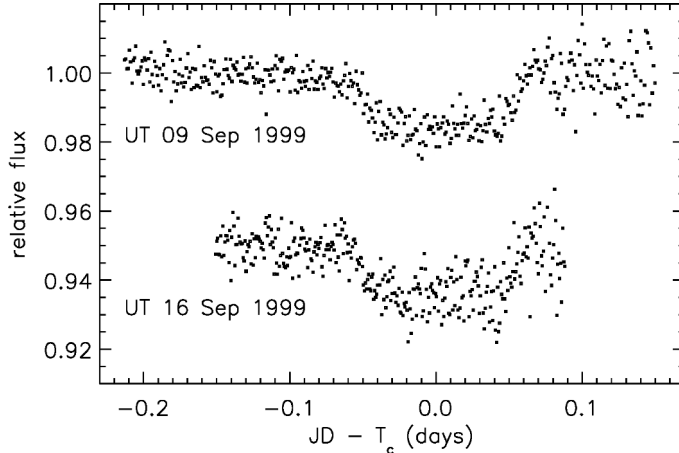


1. Historical Context

HD 209458 b first transit, STARE 1999. Charbonneau et al. (2000)

From Zero to Six Thousand in Three Decades

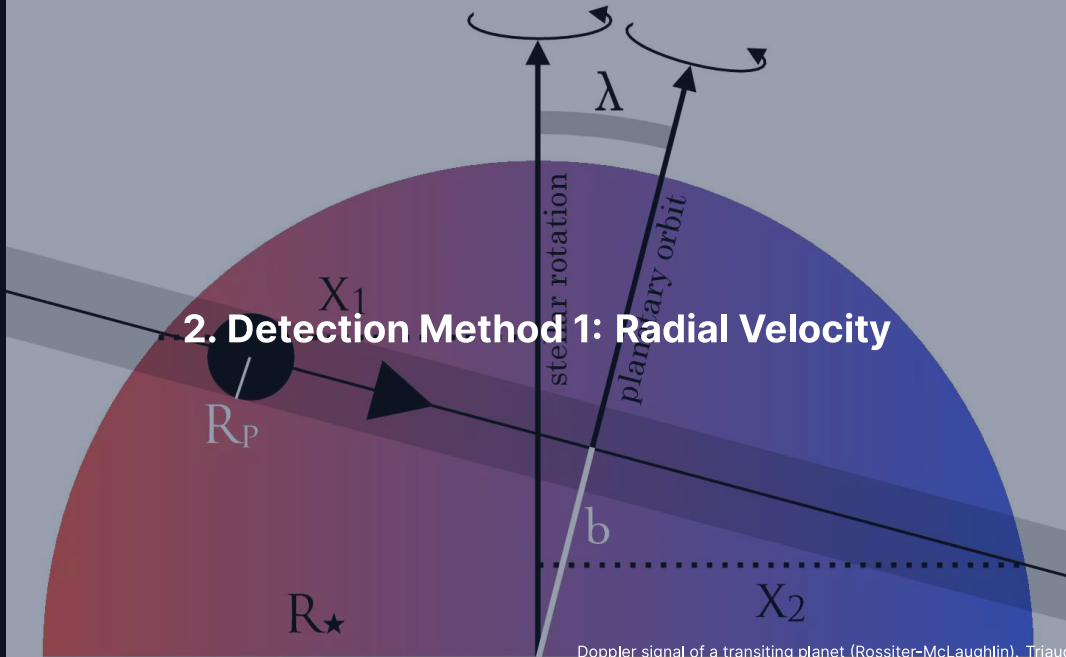
- **1992 (PSR B1257+12):** first exoplanets, around a pulsar, by timing (Wolszczan & Frail)
- **1995 (51 Pegasi b):** first planet around a Sun-like star; a 4.23-day orbit forced migration theory (Mayor & Queloz)
- **1999 (HD 209458 b):** first transit of an RV planet: a gas-giant radius and bulk density



The first ground-based detection of a transiting exoplanet, HD 209458 b, on two nights in 1999. Charbonneau et al. (2000).

Successive transits dropped the flux by about 1.7% (independent detection by Henry et al. 2000): exoplanets became physical objects with measurable radii.

2. Detection Method 1: Radial Velocity

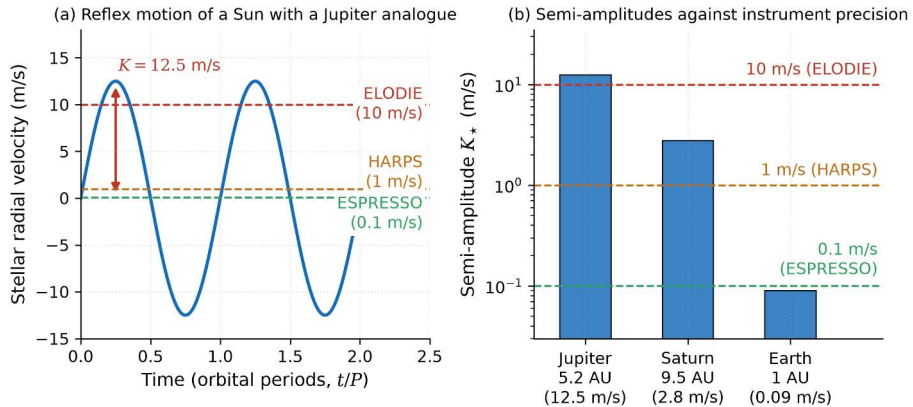


Radial Velocity: Doppler Reflex of the Star

- Star and planet orbit a common barycentre: $a_{\star} = (m_p / M_{\star}) a_p$
- The periodic Doppler shift of the stellar absorption lines gives the reflex velocity
- Semi-amplitude $K_{\star}$ for inclination i (edge-on: $\sin i = 1$; circular: $e = 0$):

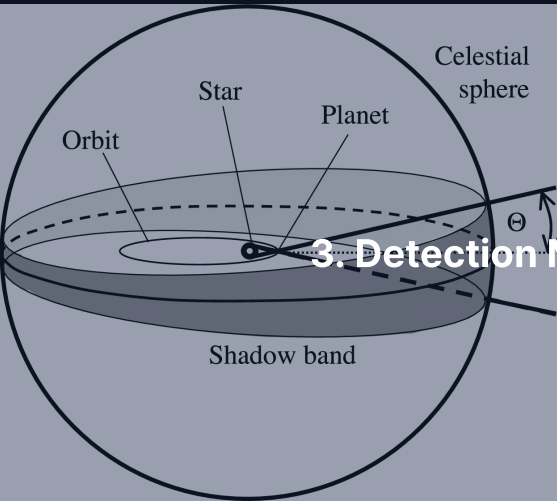
$$K_{\star} = \left(\frac{2\pi G}{P} \right)^{1/3} \frac{m_p \sin i}{M_{\star}^{2/3}} \frac{1}{\sqrt{1 - e^2}}$$

Jupiter at 5.2 AU: $K_{\star} \approx 12.5 \text{ m s}^{-1}$; Earth at 1 AU: $K_{\star} \approx 9 \text{ cm s}^{-1}$.

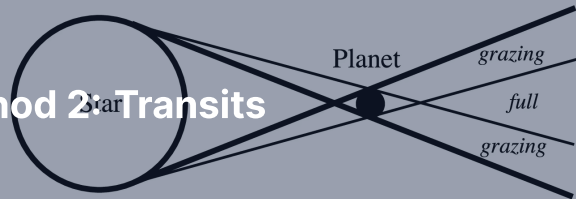


(a) Reflex velocity of a Sun with a Jupiter analogue against ELODIE, HARPS and ESPRESSO precisions; (b) semi-amplitudes of Jupiter, Saturn and Earth analogues. Course figure.

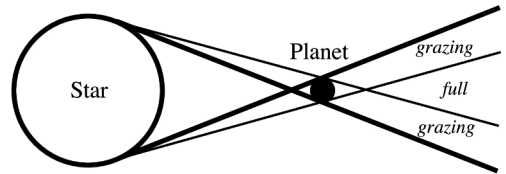
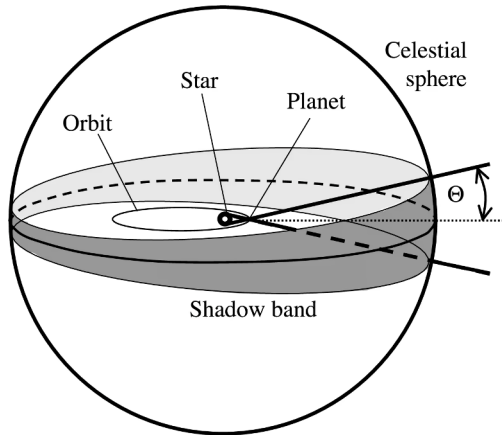
An Earth analogue at 0.09 m s^{-1} sits below the instrumental floor and within the $0.1\text{--}1 \text{ m s}^{-1}$ stellar jitter; RV alone gives only the minimum mass $m_p \sin i$.



3. Detection Method 2: Transits



Close-up



Close-up

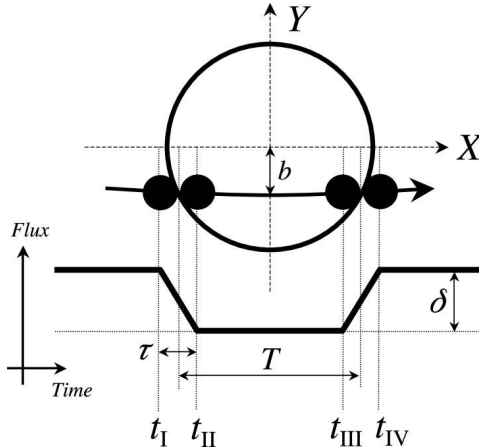
Transit geometry: a planet on an orbit near the line of sight crosses the stellar disk.
Winn (2010).

Transit depth $\delta = (R_p/R_*)^2$ and geometric probability $p \approx R_*/a$: a strong selection bias toward short periods and large R_p/R_* .

Quick Scales

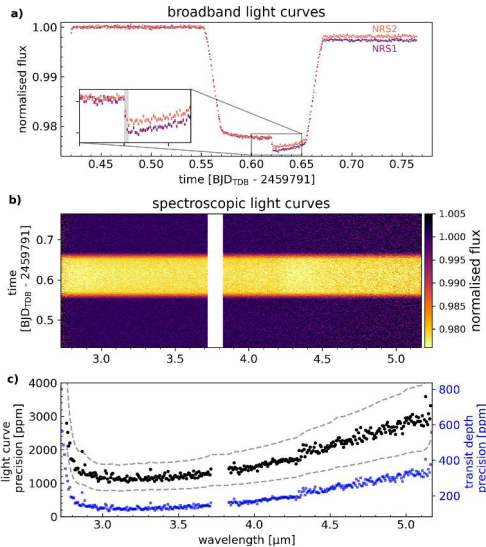
Transit depth scales inversely with the square of the stellar radius, so small planets around small stars are far easier to detect.

- **Jupiter around Sun:** $\delta \approx 1.01\%$, easily detected from ground-based telescopes
- **Earth around Sun:** $\delta \approx 84$ ppm, requires ultra-precise space photometry
- **Earth around M dwarf ($0.15 R_{\odot}$):** $\delta \approx 4000$ ppm, accessible from ground and space



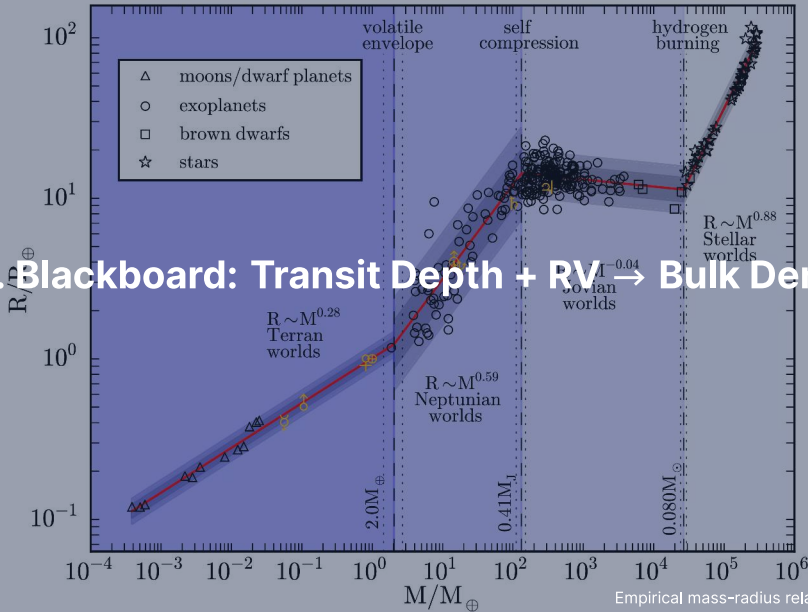
Anatomy of a transit light curve with its four contact times t_I to t_{IV} . Winn (2010).

The depth gives the radius ratio R_p/R_* ; the total duration and the ingress time give the impact parameter b and the inclination. A real transit bottom curves from stellar limb darkening.

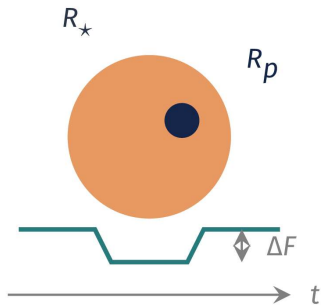


JWST/NIRSpec transit photometry of WASP-39 b: depths measured to 200–600 ppm per spectroscopic bin, below typical exoplanet absorption features. Alderson et al. (2023).

4. Blackboard: Transit Depth + RV $\rightarrow$ Bulk Density



Goal and strategy



Goal: a radius and a mass for the same planet, hence a density.

- Transit alone: the radius R_p , but no mass
- RV alone: the minimum mass $m_p \sin i$, but no radius
- Combined: **both**, so the bulk density $\bar{\rho}_p$

Bulk density turns an astronomical detection into a physical, geological body. To the board.

Steps 1 to 3 and the Density

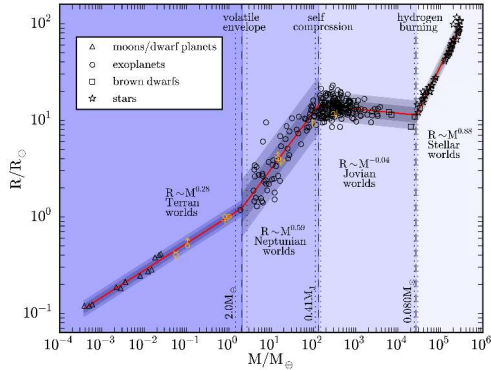
Transit depth $\delta = (R_p/R_\star)^2$ gives R_p ; the reflex velocity

$K_\star = (2\pi G/P)^{1/3} m_p \sin i M_\star^{-2/3}$ gives $m_p \sin i$ (broad).

- A transit forces $\sin i \approx 1$ and collapses the $m_p \sin i$ degeneracy
- Both R_p and m_p measured for the same object
- Bulk density follows from elementary geometry:

$$\bar{\rho}_p = \frac{m_p}{(4/3)\pi R_p^3} = \frac{3m_p}{4\pi R_p^3}$$

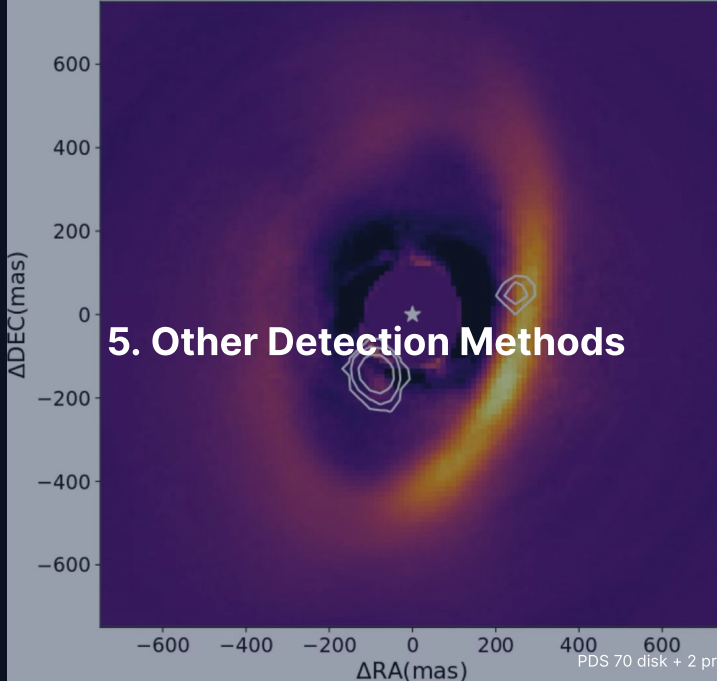
Transit plus RV turns a detection into a physical object: radius, mass and density from one planet.



The empirical mass-radius relation of exoplanets and solar-system bodies.
Chen & Kipping (2017).

Density maps to composition: $\bar{\rho} \approx 5.5 \text{ g cm}^{-3}$ rocky (Earth, Venus), 1.3 g cm^{-3} gas giant (Jupiter), 0.5 g cm^{-3} an inflated envelope, and $1\text{--}3 \text{ g cm}^{-3}$ the sub-Neptunes, a size class the solar system lacks.

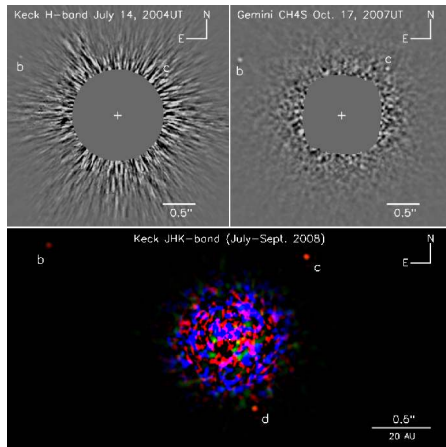
5. Other Detection Methods



PDS 70 disk + 2 protoplanets. Haffert et al. (2019)

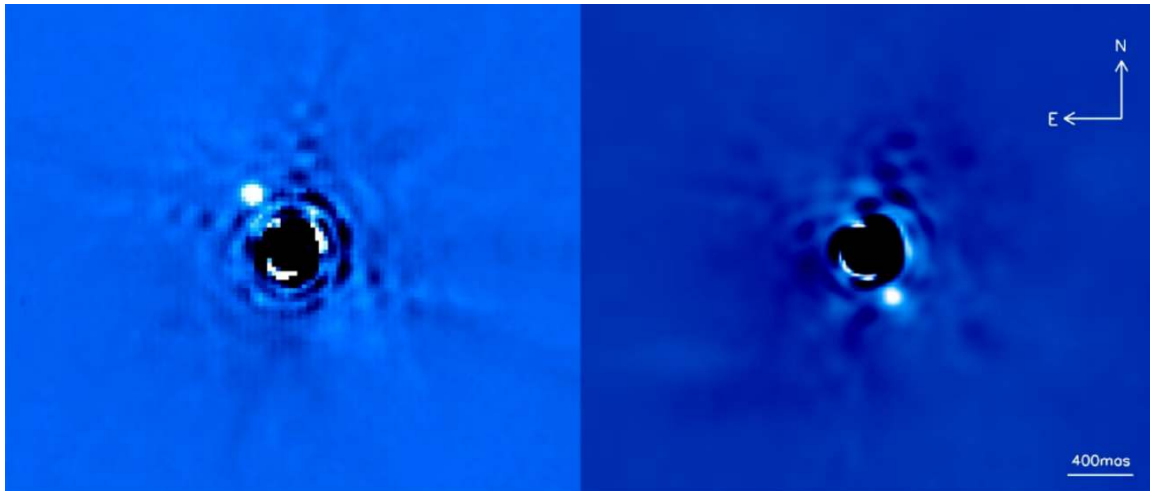
Direct Imaging: Spatial Separation of Photons

- **Contrast:** suppress the star by 10^{-9} (Jupiter) to 10^{-10} (Earth twin) at separation $\theta = a/d$
- **Technology:** extreme adaptive optics plus coronagraphs against diffraction and speckle noise
- **Bias:** young, massive, self-luminous giants on wide orbits ($a > 10$ AU)

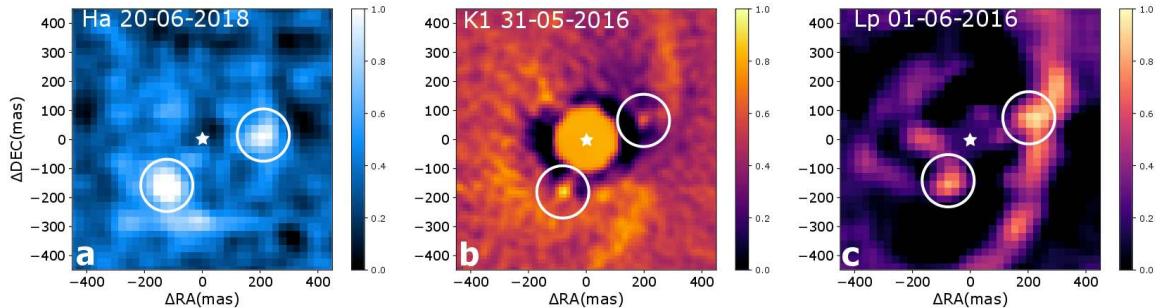


Keck and Gemini adaptive-optics discovery images of the HR 8799 planets.
Marois et al. (2008).

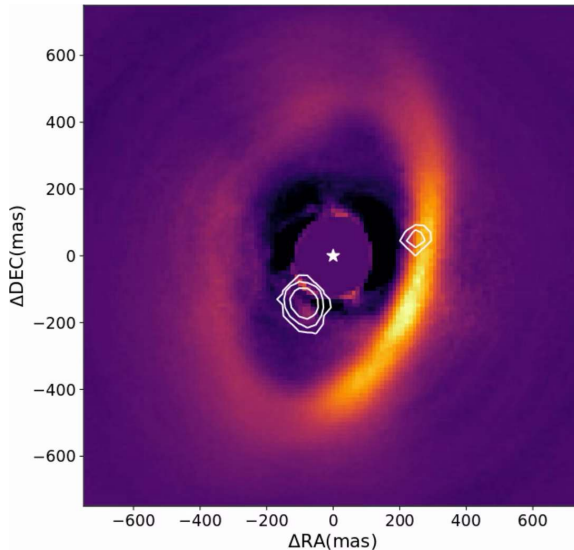
Four co-moving planets of 5–10 M_J at 14–68 AU (planet e added in 2010) around a 30–60 Myr star: they still glow from their contraction heat.



Direct image of β Pictoris b: orbital motion between 2003 and 2009 confirms a bound $\sim 9 M_J$ companion at ~ 9 AU. Lagrange et al. (2010); credit: ESO/A.-M. Lagrange et al.



PDS 70 b and c inside the disk gap in MUSE H α (a), SPHERE K1 (b) and NACO L' (c): both protoplanets are accreting within the cleared cavity. Haffert et al. (2019).



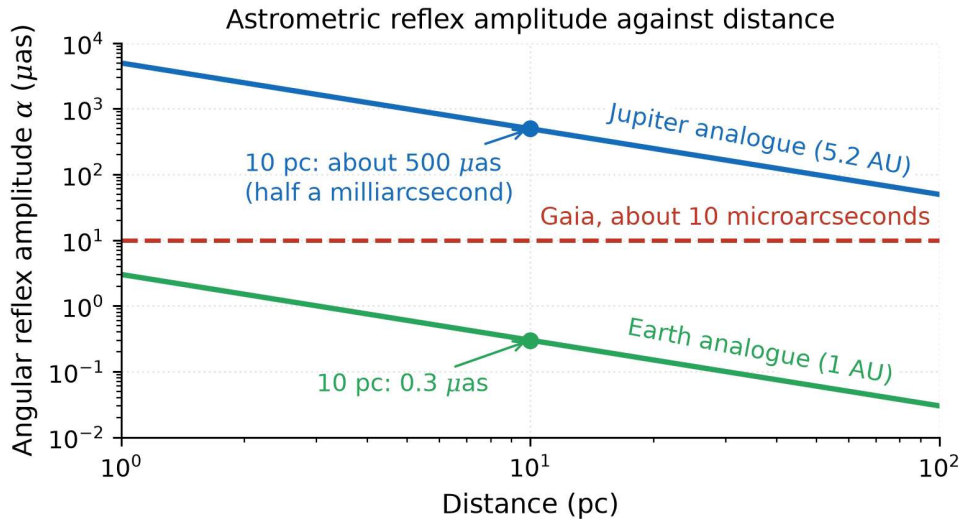
Composite scattered-light image of the PDS 70 protoplanetary disk with the two embedded protoplanets inside the cleared gap. Haffert et al. (2019).

Astrometry: Gaia Will Find Jupiter Analogues

Astrometry follows the star's reflex wobble on the sky and gives true masses.

$$\alpha = \frac{m_p}{M_\star} \cdot \frac{a_p}{d}$$

- **Signal:** a Jupiter analogue at 10 pc gives $\alpha \sim 500 \mu\text{as}$; an Earth analogue $\sim 0.3 \mu\text{as}$
- **Gaia DR4 (2026):** thousands of cold giants expected; $\sim 10 \mu\text{as}$ final precision
- **True mass:** the inclination i is measured, so no $m \sin i$ degeneracy



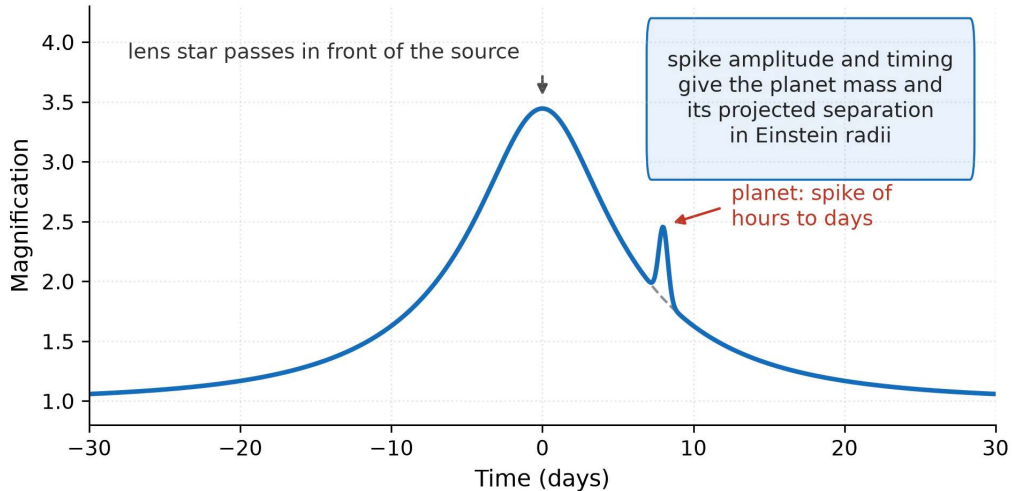
Astrometric reflex amplitude against distance for Jupiter and Earth analogues, with Gaia's final precision of about 10 microarcseconds as the dashed line. Course figure.

Microlensing, Timing, and Detection Biases

- **Microlensing (Roman):** cool planets at 1–10 AU and unbound rogue worlds
- **Timing variations (TTVs):** dynamical masses for faint hosts such as TRAPPIST-1
- **Together:** transits and RV probe close-in orbits; imaging, astrometry and microlensing the wide ones

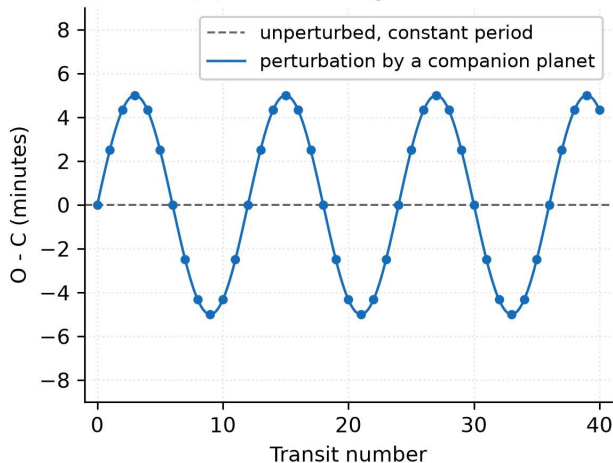
Each method covers its own part of parameter space, so the archive carries the biases of the methods.

A microlensing event with a planet



A microlensing event with a planet: the magnification of a background source by a lens star, and the spike of hours to days that a planet around the lens adds. Course figure.

(a) Transit timing variations



(b) Timing methods

transit timing variations:

dynamical masses (TRAPPIST-1)

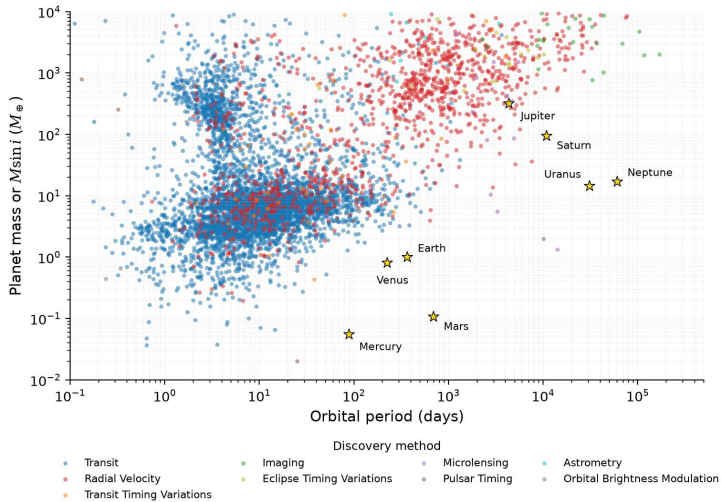
pulsar timing:

the first exoplanets (1992)

eclipse timing in binaries:

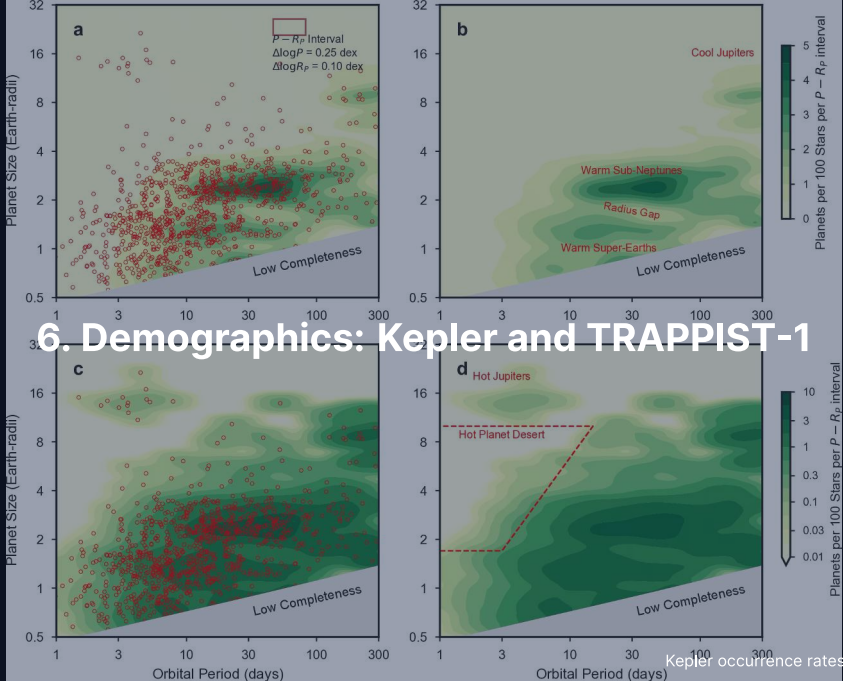
circumbinary planets
(Kepler-16 b, 2011)

(a) Observed minus calculated transit times of a planet perturbed by a companion (5 minute amplitude, schematic); (b) the three timing methods: transit timing, pulsar timing and eclipse timing in binaries. Course figure.



Confirmed exoplanet masses (or $m_p \sin i$) against orbital period by detection technique, with the solar-system planets: discoveries pile up at high mass and short period.

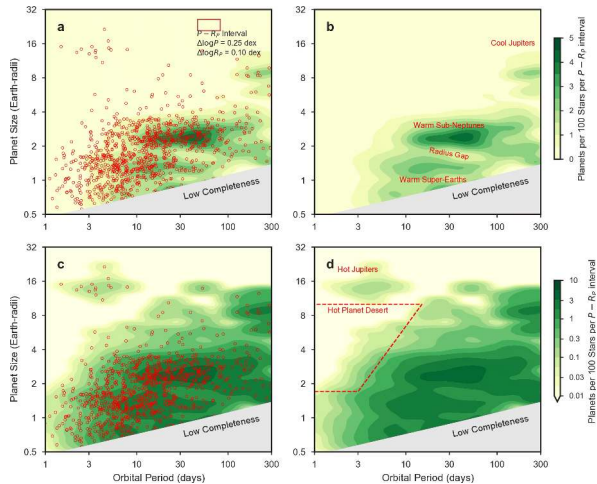
Credit: NASA Exoplanet Archive (2026). Course figure.



The Kepler Revolution

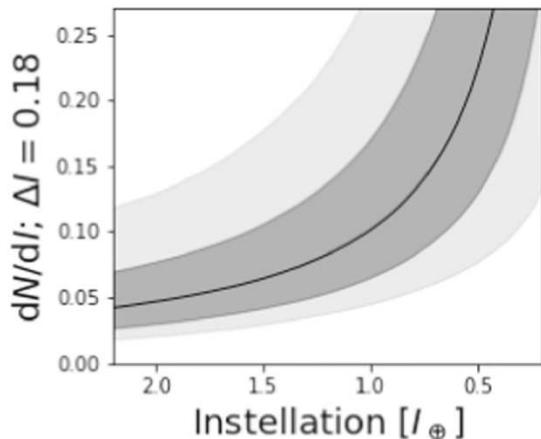
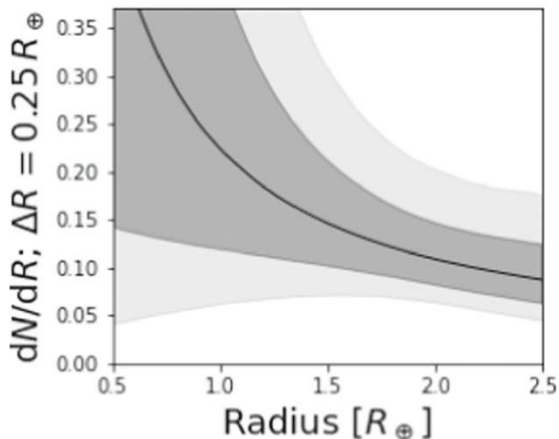
- **Planets are ubiquitous:** on average more than 1 planet per main-sequence star (Petigura et al. 2018)
- **Small planets dominate:** super-Earths and sub-Neptunes are common; hot Jupiters are rare (0.5–1%)
- **Habitable terrestrial rate:** Earth-sized planets in the habitable zone occur at $\eta_{\oplus} \sim 0.4$ (Bryson et al. 2021)

Kepler's computable detection efficiency gave the first bias-corrected occurrence rates: planets outnumber stars in the Milky Way.



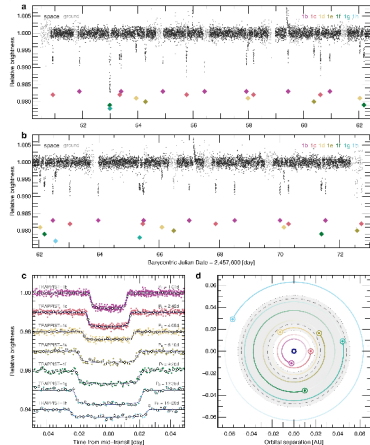
Planet occurrence against period and size, California-Kepler Survey. Petigura et al. (2018).

Small planets outnumber giants at every period; the fall-off beyond ~ 100 days is partly completeness-limited.



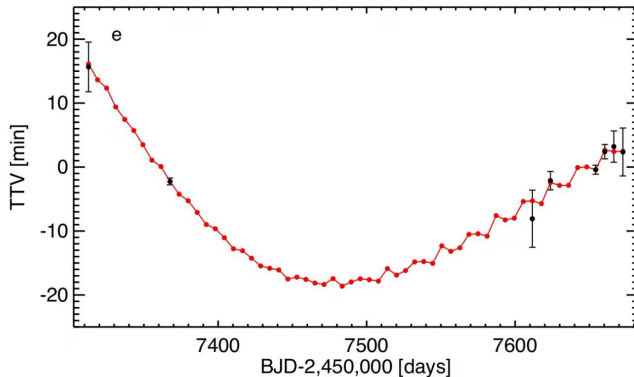
Marginalised occurrence of small planets per unit radius (left) and per unit instellation (right), with 68% and 95% credible bands. Bryson et al. (2021).

The central estimate $\eta_{\oplus} \sim 0.4$: terrestrial planets in habitable zones are common.



Transits of the seven TRAPPIST-1 planets. Gillon et al. (2017).

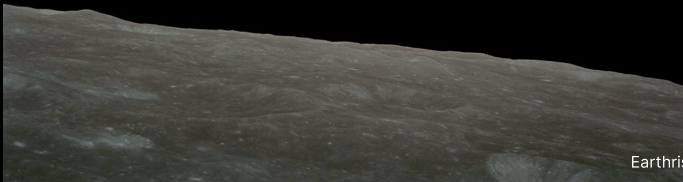
An M8 ultra-cool dwarf at 12 pc with seven Earth-sized planets within about 0.06 AU in a resonant chain, consistent with early inward migration; e, f and g orbit in the temperate habitable zone.



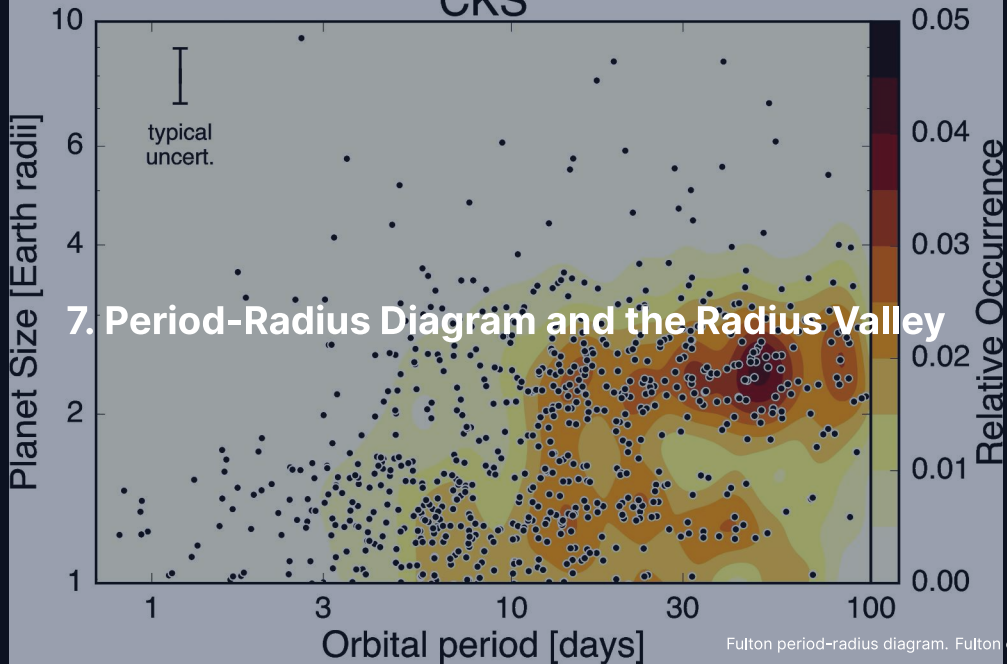
Transit-timing variations of TRAPPIST-1 e against a seven-planet dynamical model.
Gillon et al. (2017).

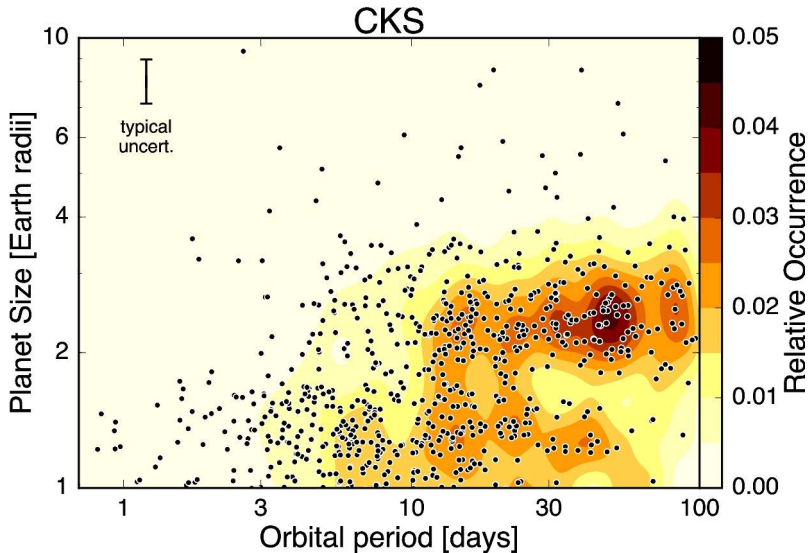
Timing variations of tens of minutes, fitted with simultaneous N-body models, break the mass degeneracies and give individual masses and densities without radial velocities.

Break

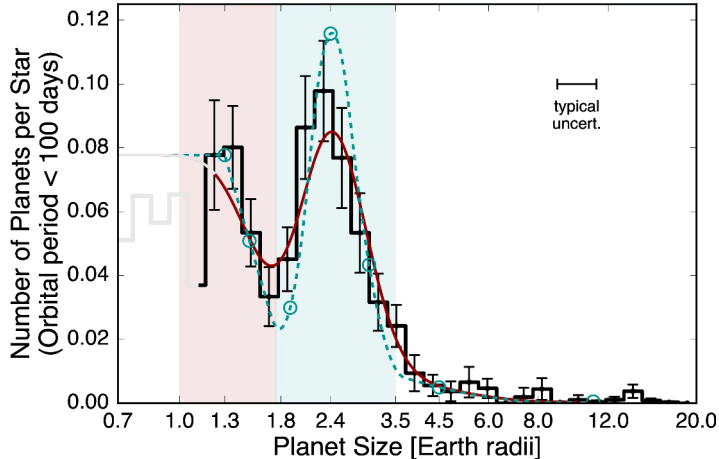


CKS





Period-radius distribution of small Kepler planets after stellar parameter refinement: super-Earths and sub-Neptunes separated by a deficit at $\sim 1.8 R_{\oplus}$. Fulton et al. (2017).



The radius valley: small Kepler planets have a bimodal size distribution. Fulton et al. (2017).

A deficit at $1.5\text{--}2.0 R_{\oplus}$ separates super-Earths near $1.3 R_{\oplus}$ (bare rocky cores) from sub-Neptunes near $2.4 R_{\oplus}$ (volatile envelopes); photoevaporation or core-powered mass loss carves it.

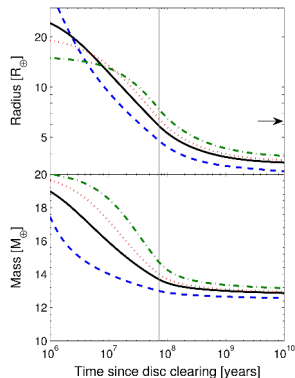
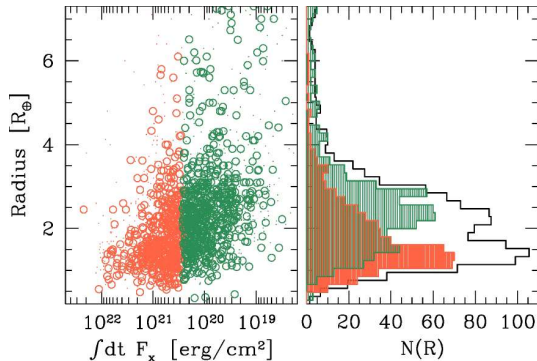
Photoevaporation Strips H/He Envelopes

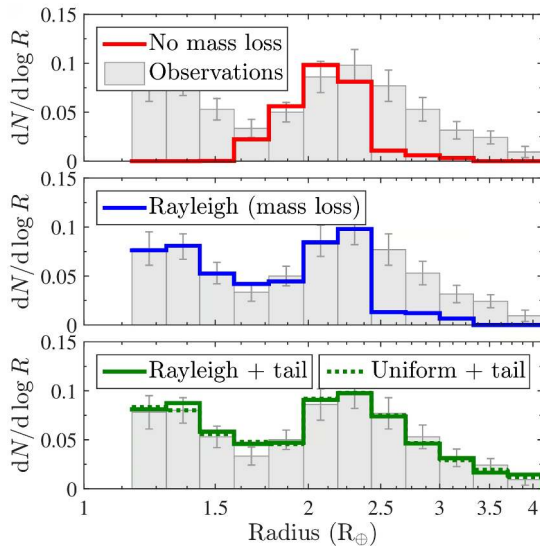
Energy-limited hydrodynamic escape driven by high stellar X-ray and EUV irradiation:

$$\dot{M} \approx \frac{\epsilon \pi F_{\text{XUV}} R_p^3}{GM_p}$$

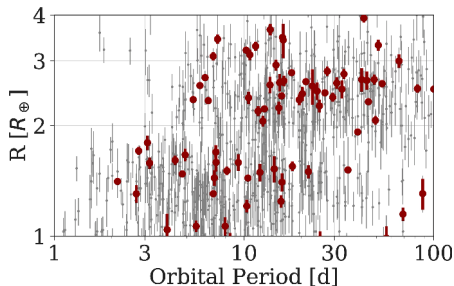
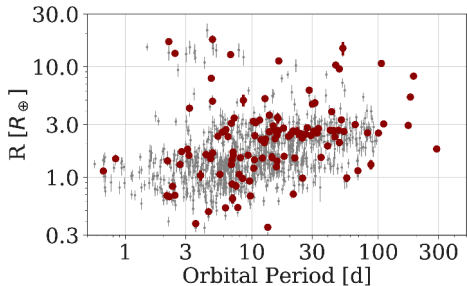
- Young stars saturate XUV flux at $\sim 10^3 \times$ modern levels for ~ 100 Myr
- Strips light primordial H/He envelopes from lower-mass cores at close-in orbits
- Owen & Wu (2013): naturally carves the bimodal radius valley

Photoevaporation Model + Mass-Loss Tracks



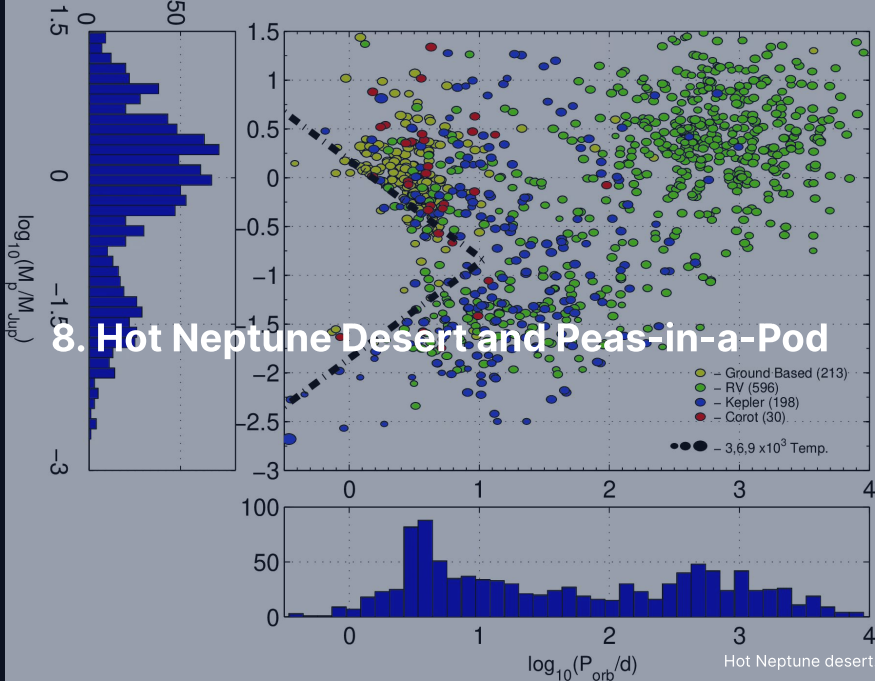


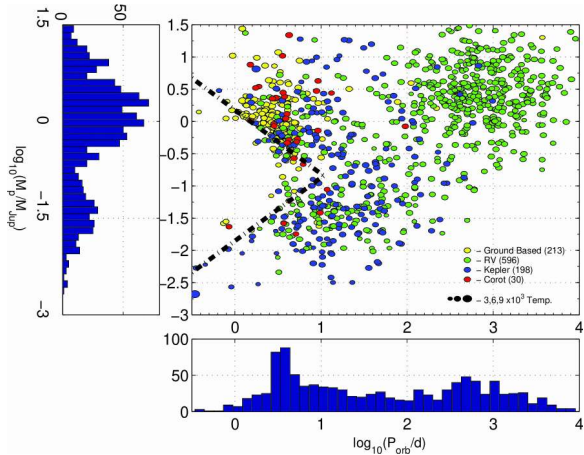
Core-powered mass loss: envelope loss driven by heat from the cooling rocky interior also produces a bimodal distribution with a gap at $\sim 1.8 R_{\oplus}$. Ginzburg et al. (2018).



The radius valley with asteroseismic stellar radii: its position shifts with orbital period. Van Eylen et al. (2018).

The valley slopes to smaller R_p at longer periods, as photoevaporation and core-powered cooling predict: many close-in super-Earths are stripped cores rather than primordial rocky worlds.

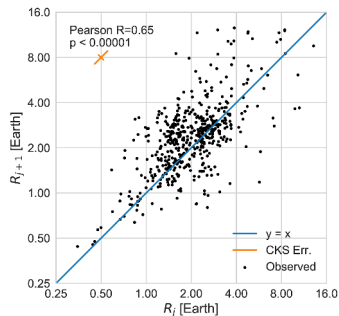




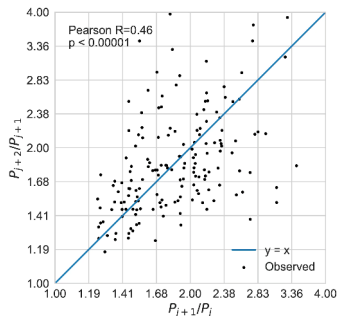
The hot Neptune desert in the mass-period plane. Mazeh et al. (2016).

A deficit of $10\text{--}100 M_{\oplus}$ planets at $P < 5$ d: at the **bottom edge** photoevaporation strips low-mass envelopes and leaves super-Earths, at the **top edge** tidal Roche-lobe overflow strips inflated giants.

Peas-in-a-Pod Architectures



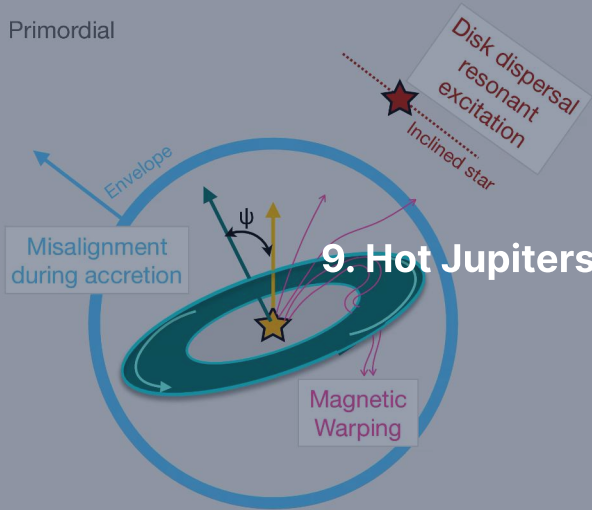
Adjacent radii: Pearson $r = 0.65$



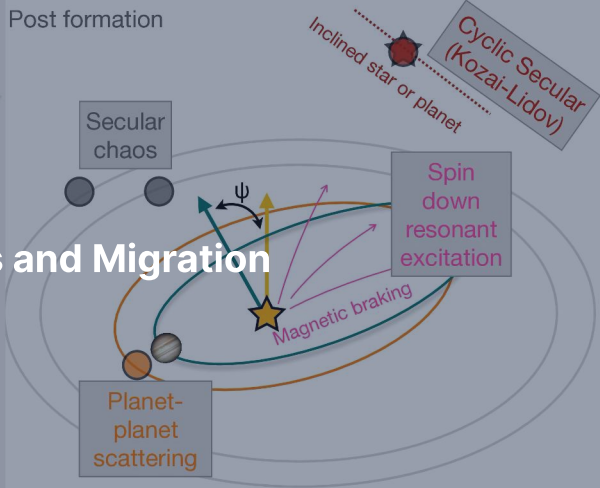
Adjacent period ratios: clustered

- Planets in compact multi-systems tend to have similar sizes and uniform orbital spacing (Weiss et al. 2018)
- Favours quiet in-situ growth or migration over chaotic giant impacts

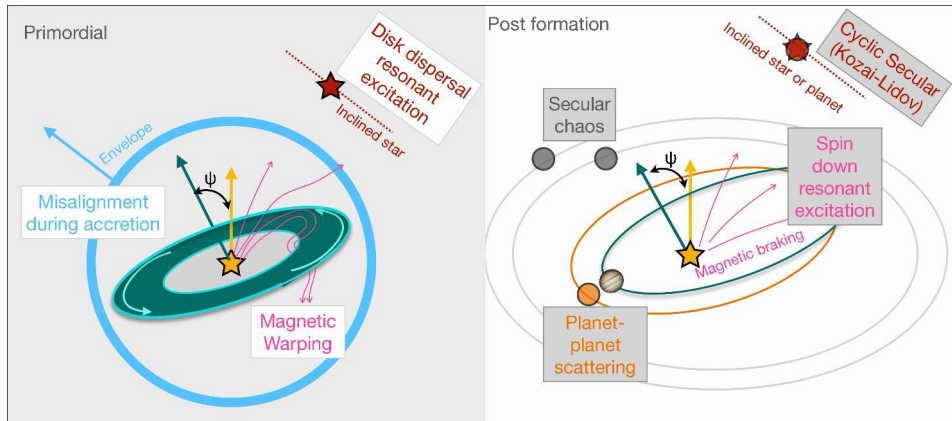
Primordial



Post formation

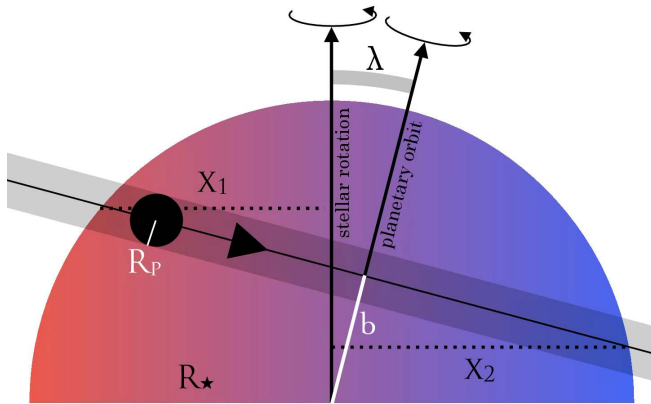


9. Hot Jupiters and Migration



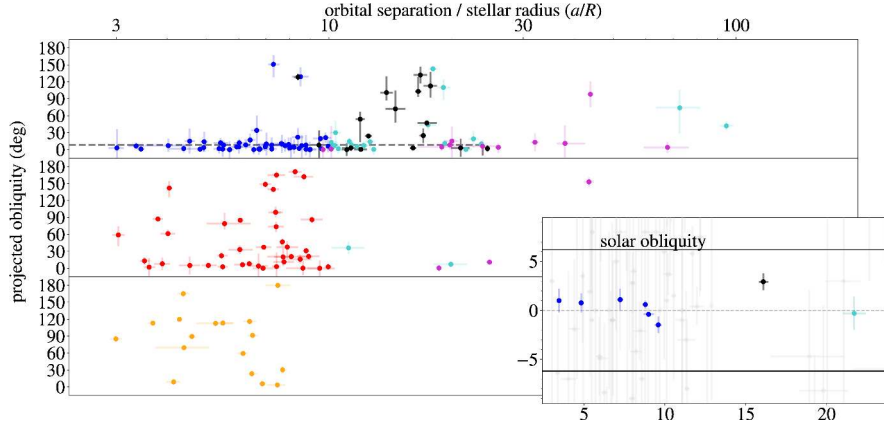
Three pathways to a hot Jupiter and the stellar obliquity each one leaves.
Albrecht et al. (2022).

Disk migration (Type II) keeps a low obliquity; **high-eccentricity** Kozai-Lidov migration and **planet scattering** leave wide misalignments; in situ growth at 0.05 AU is disfavoured.



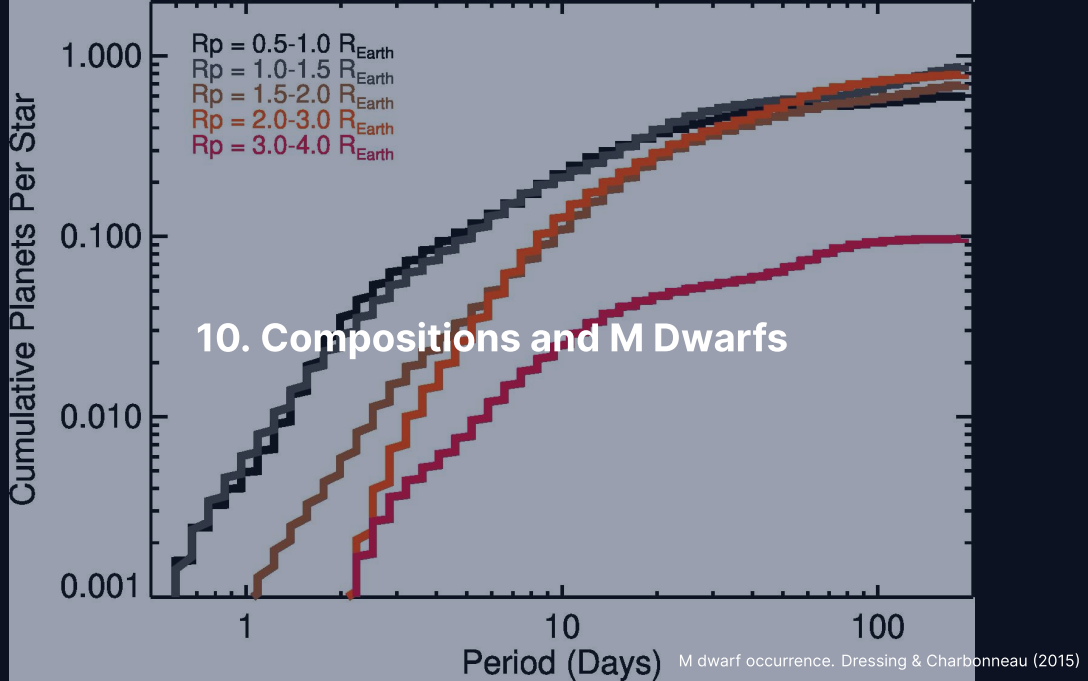
The Rossiter-McLaughlin effect: the line-profile distortion from a planet crossing a rotating star. Triaud (2018).

On an aligned orbit the planet first hides the approaching, then the receding hemisphere; the time-resolved line distortion gives the sky-projected spin-orbit angle λ , which separates disk migration from dynamical scattering.



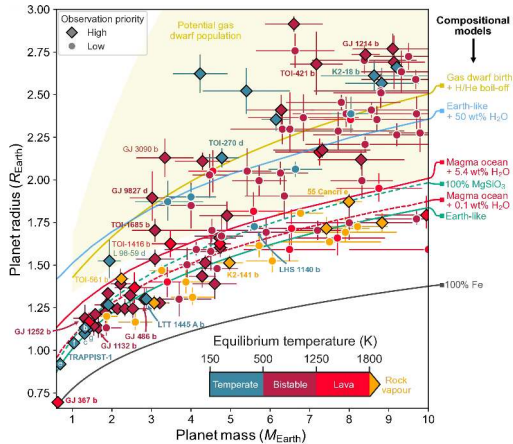
Stellar obliquities of hot Jupiter hosts against a/R_* , by effective temperature.
Albrecht et al. (2022).

Cool stars ($T_{\text{eff}} < 6250$ K) realign tidally through their convective envelopes; **hot stars** keep high obliquities; the wide dispersion at large a/R_* shows that dynamical migration occurs.



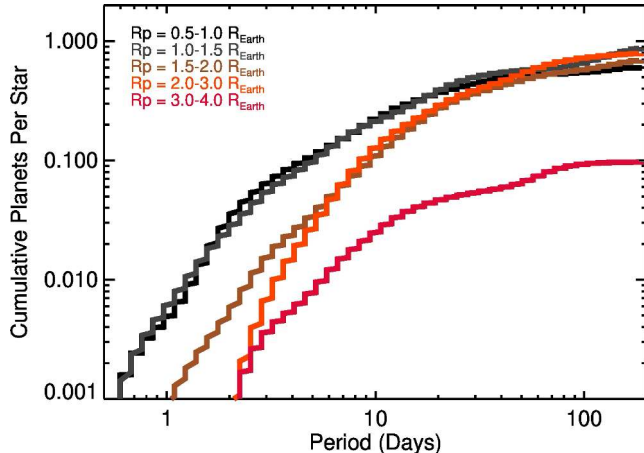
Sub-Neptunes: Water World or H/He?

- **Density split:** super-Earths ($\bar{\rho} \sim 4\text{--}8 \text{ g cm}^{-3}$) are rocky; sub-Neptunes ($\bar{\rho} \sim 1\text{--}3 \text{ g cm}^{-3}$) require volatile envelopes
- **Degeneracy:** density cannot separate a 1% H/He envelope from a 50% water-rich mantle
- **Spectroscopic test:** transmission spectroscopy gives the atmospheric mean molecular weight



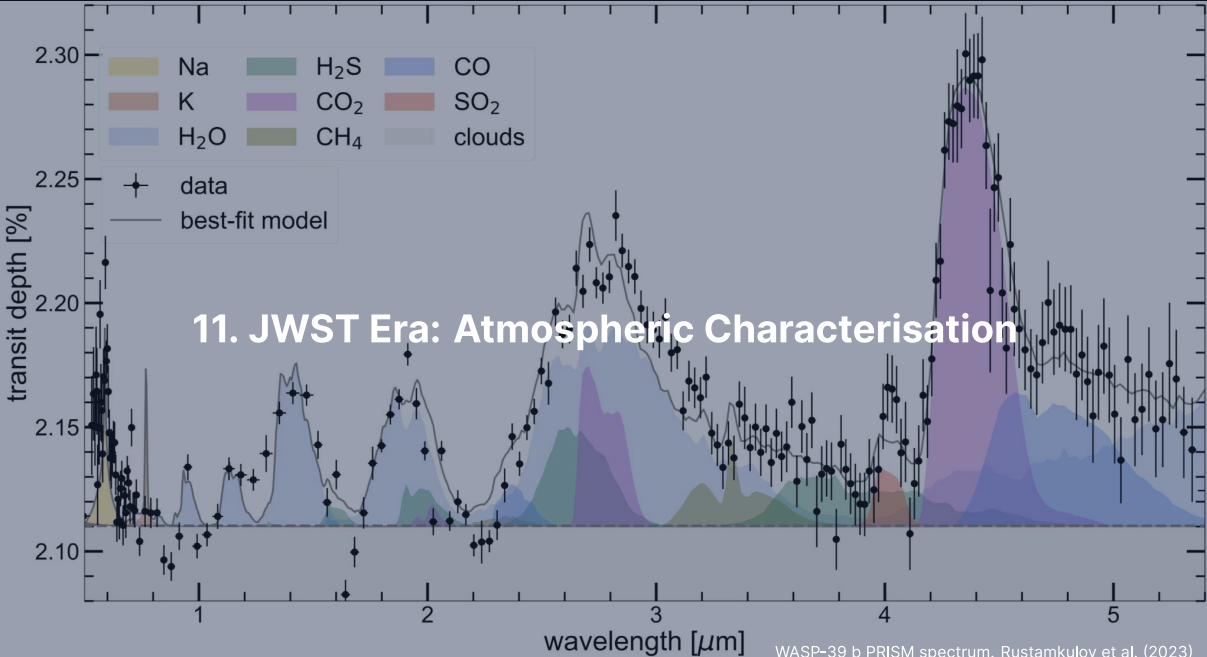
Mass against radius for small exoplanets with interior models from pure iron to volatile-rich envelopes, sorted by thermal regime. Lichtenberg et al. (2025).

Rocky super-Earths follow the silicate-iron curves; sub-Neptunes need substantial volatile envelopes or water-rich layers to explain their larger radii.



Planet occurrence around M dwarfs from Kepler. Dressing & Charbonneau (2015).

M dwarfs host ~ 2.5 small planets each and favour transits and RV, but **XUV desiccation** in the long pre-main sequence can strip whole oceans (Luger & Barnes 2015) and **tidal locking** fixes day and night sides.

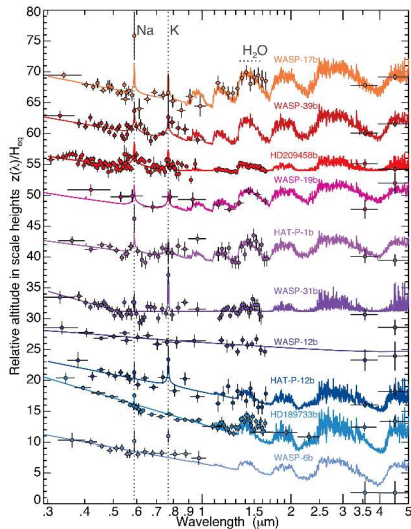


Transmission Spectroscopy

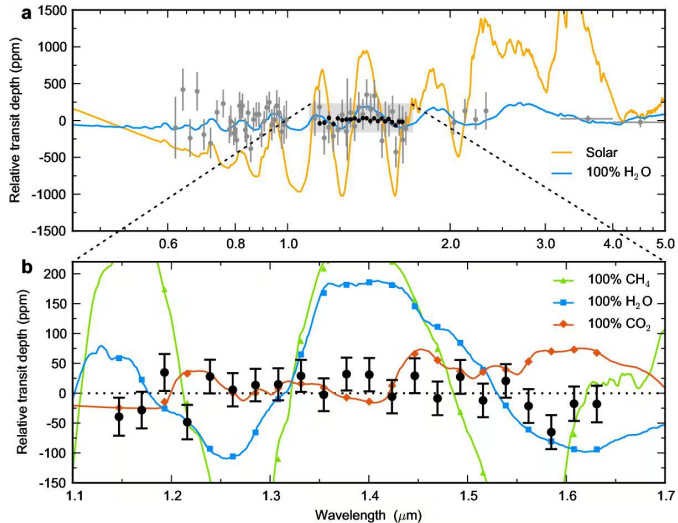
Starlight filtered through an atmosphere during transit encodes molecular absorption; with the scale height

$$H = \frac{k_B T}{\mu m_u g}, \quad \delta(\lambda) \approx \frac{(R_p + n_H(\lambda) H)^2}{R_\star^2}, \quad \frac{\Delta\delta}{\delta} \approx \frac{2n_H H}{R_p}.$$

- Hot Jupiter ($T = 1500$ K, $\mu = 2.3$, $g = 25 \text{ m s}^{-2}$): $H \approx 200$ km, $\Delta\delta/\delta \approx 2\%$, absolute modulation $\Delta\delta \sim 200$ ppm (JWST-routine)
- Sub-Neptunes: 10–100 ppm; terrestrial atmospheres: < 10 ppm
- High clouds or hazes flatten the spectrum (GJ 1214 b): a flat spectrum does not mean no atmosphere

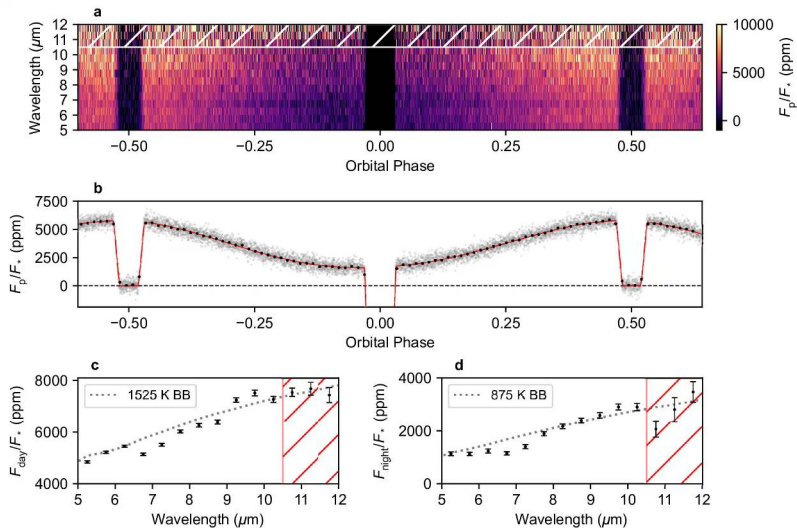


Transmission spectra of ten hot Jupiters from HST and Spitzer across $0.3\text{--}5\ \mu\text{m}$: a continuum from clear atmospheres with water absorption to cloudy scattering slopes. Sing et al. (2016).

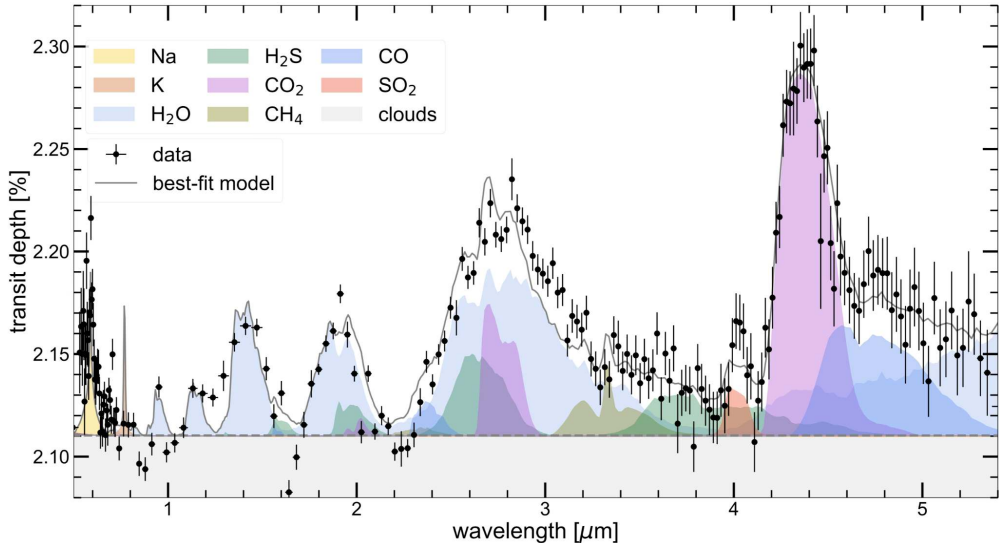


Featureless HST transmission spectrum of the warm sub-Neptune GJ 1214 b: cloud-free H₂O, CH₄ and CO₂ atmospheres are ruled out, so high clouds or hazes are present.

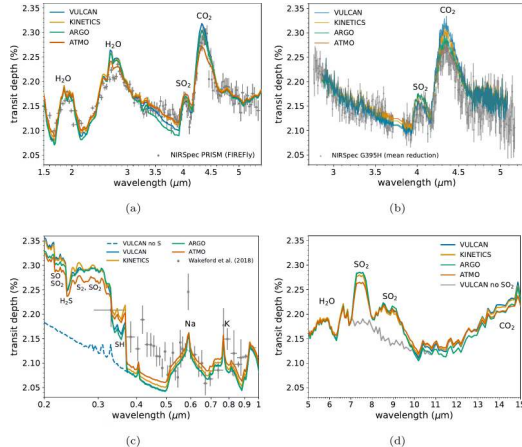
Kreidberg et al. (2014).



JWST/MIRI spectroscopic phase curve of WASP-43 b: thermal emission from dayside to nightside constrains the atmospheric heat redistribution. Bell et al. (2024).



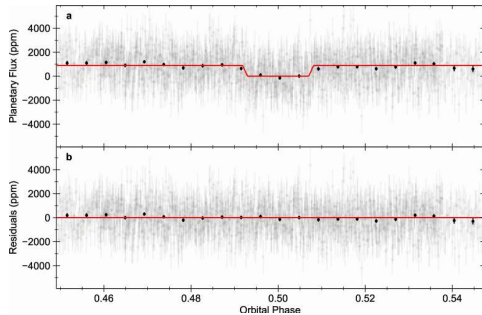
JWST/NIRSpec PRISM transmission spectrum of WASP-39 b with model opacity bands: H₂O (33 σ), CO₂ (28 σ), clouds (21 σ), Na (19 σ) and CO (7 σ). Rustamkulov et al. (2023).



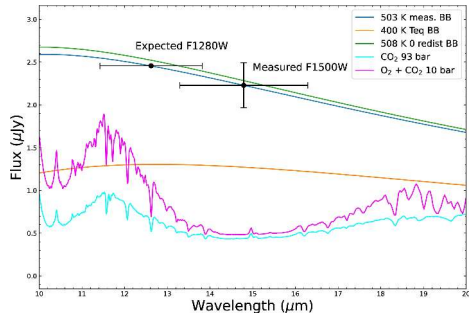
The $4.05 \mu\text{m}$ absorption feature of SO_2 in the JWST spectrum of the hot Saturn WASP-39 b ($0.28 M_J$). Tsai et al. (2023).

Photochemical SO_2 from UV photolysis of H_2S : the first photochemical product identified in an exoplanet atmosphere, proof of disequilibrium chemistry.

TRAPPIST-1 b: Bare Rock Dayside

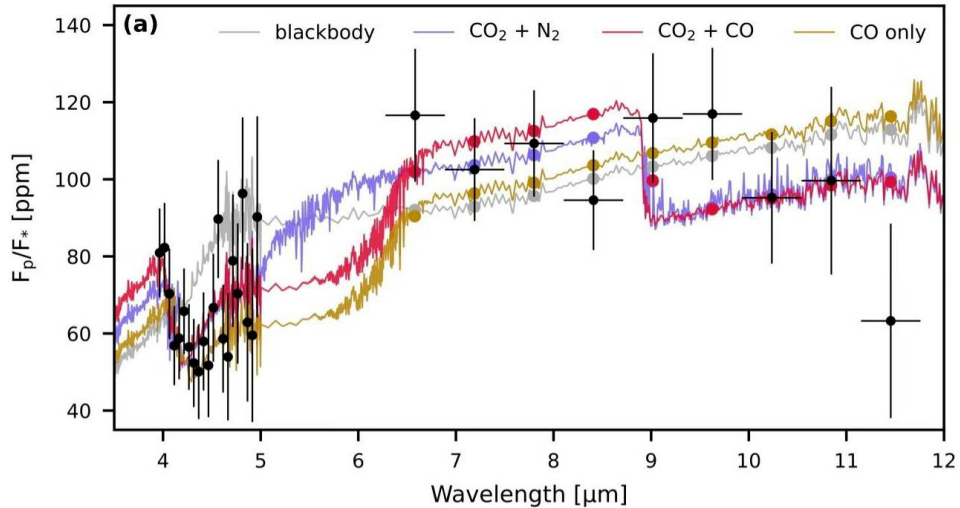


15 μm secondary eclipse

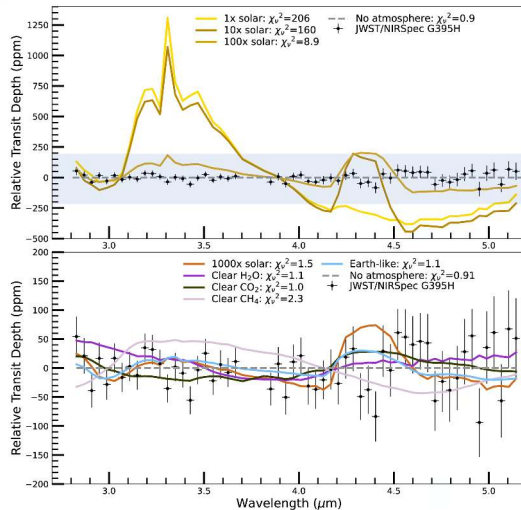


Compared to model atmospheres

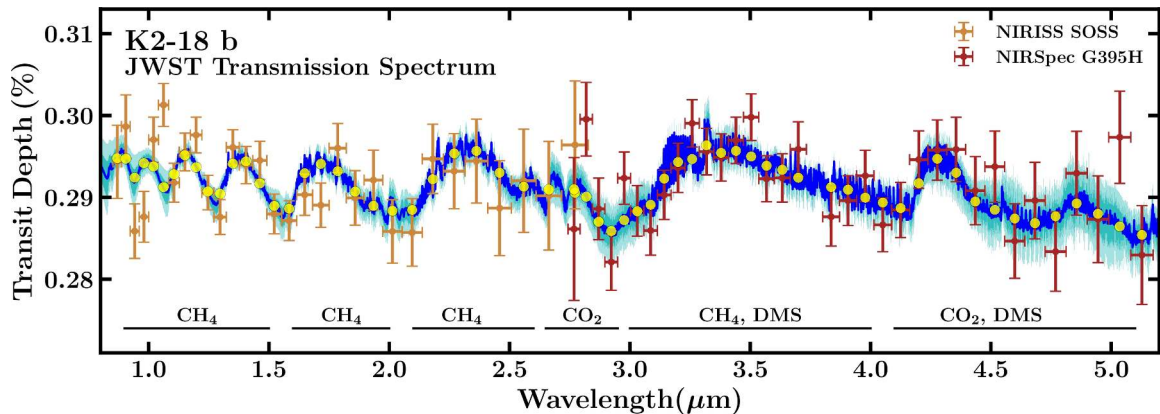
$T_B = 503^{+26}_{-27}$ K, consistent with bare rock; a **thick CO₂ atmosphere is ruled out** (Greene et al. 2023). With TRAPPIST-1 c and LHS 475 b, inner M dwarf rocky planets show a widespread lack of thick atmospheres.



JWST NIRCam and MIRI emission spectrum of 55 Cancri e: the data fall below the bare-rock blackbody and favour CO and CO_2 , tentative evidence for a secondary atmosphere outgassed from a molten surface. Hu et al. (2024).

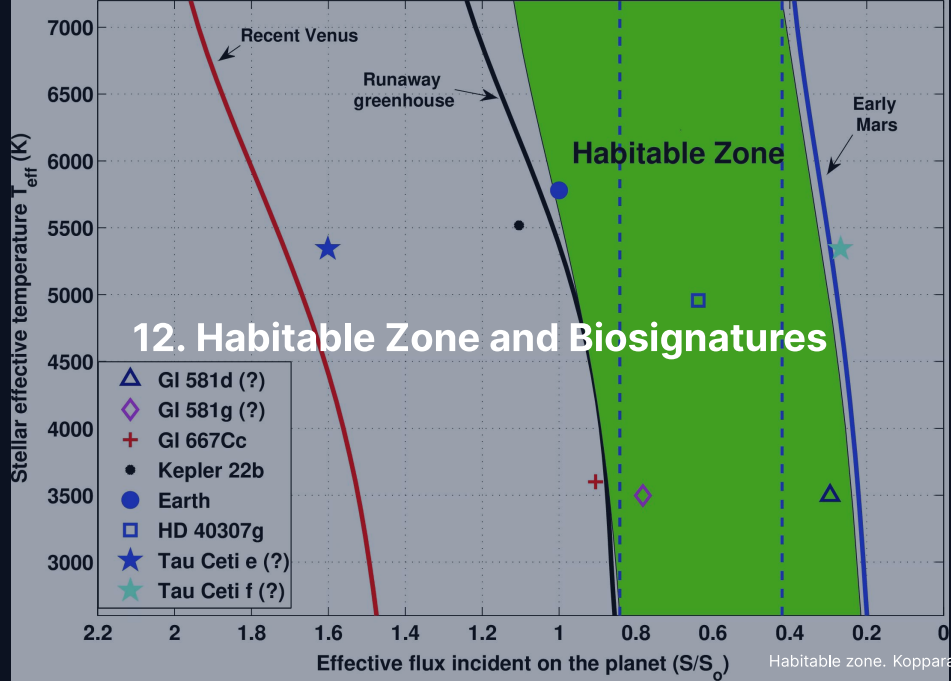


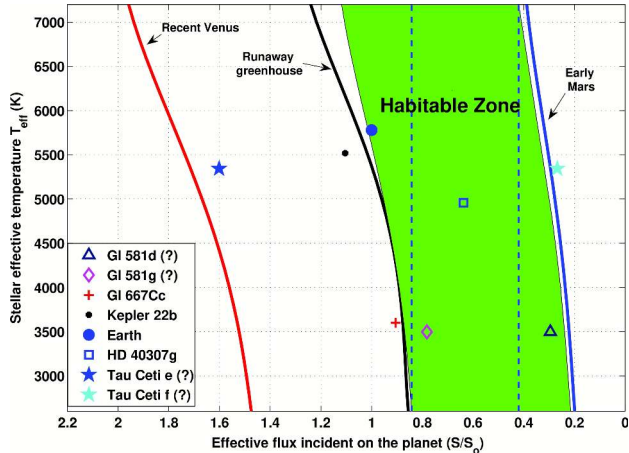
JWST NIRSpec G395H transmission spectrum of the Earth-size M dwarf planet LHS 475 b at about 12 pc: flat, so hydrogen-helium and pure CH_4 atmospheres are ruled out and a Venus-like CO_2 atmosphere stays marginally consistent. Lustig-Yaeger et al. (2023).



JWST transmission spectrum of the sub-Neptune K2-18 b in the habitable zone of its M dwarf. Madhusudhan et al. (2023).

CH_4 and CO_2 detected, DMS disputed and tentative ($\lesssim 2\sigma$); Hycean ocean, mini-Neptune envelope and magma ocean models compete: candidate biosignatures need rigorous real-time vetting.





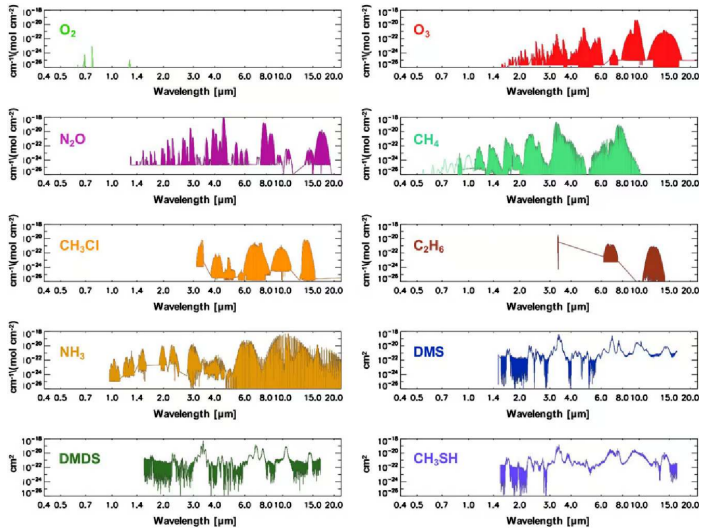
The classical habitable zone from a 1D radiative-convective grid. Kopparapu et al. (2013).

Inner edge: the moist greenhouse limit, just outside the runaway greenhouse;
 outer edge: the maximum CO_2 greenhouse before CO_2 condenses; both scale with
 stellar effective temperature and luminosity.

Habitable Zone Evolution and Biosignatures

- **Dynamic trajectory:** HZ boundaries shift with stellar evolution; pre-MS M dwarfs subject planets to prolonged runaway desiccation
- **Disequilibrium pairs:** atmospheric biosignatures demand chemical disequilibrium ($\text{O}_2 + \text{CH}_4$), not lone gas species
- **False positives:** abiotic O_2 can build up by H_2O photolysis with hydrogen escape, or by CO_2 photolysis

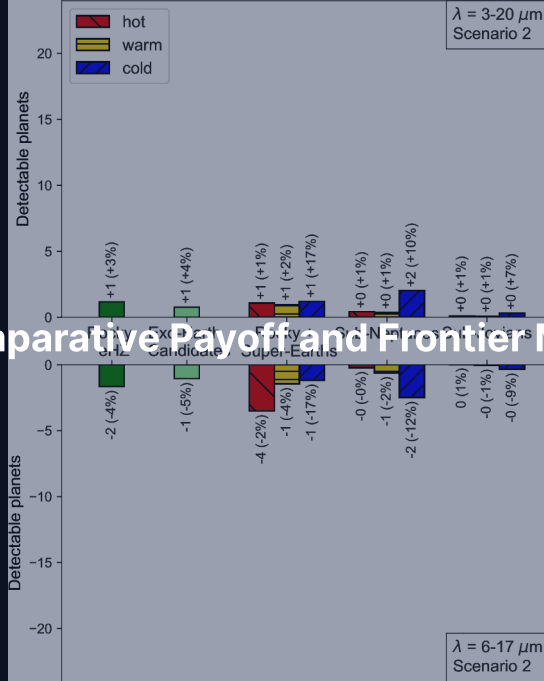
Life detection is an inverse problem requiring exclusion of all abiotic pathways.

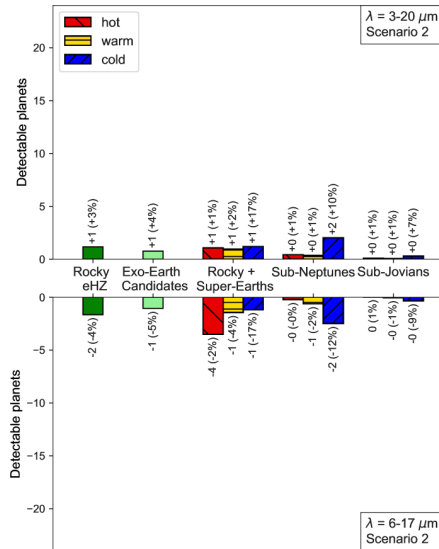


Absorption cross-sections from 0.4 to 20 μm for ten candidate biosignature gases: oxygen absorbs only in narrow optical bands, the others mostly in the infrared.

Schwieterman et al. (2018).

13. Comparative Payoff and Frontier Missions





Predicted detection yields of the LIFE mid-infrared space interferometer for exoplanets within 20 pc, by radius and insolation. Quanz et al. (2022).

Summary: Today's Course-Level Takeaways

1. **Detection:** transit plus radial velocity breaks $m \sin i$ and yields the true bulk density; every method carries its own bias
2. **Demographics:** small planets dominate ($\eta_{\oplus} \sim 0.4$); photoevaporation or core-powered mass loss carves the radius valley at $\sim 1.8 R_{\oplus}$; resonant chains and peas-in-a-pod systems record smooth disk migration
3. **Characterisation:** JWST detects photochemistry and bare rocks; biosignatures require ruling out false positives

Next: Lecture 14



Questions?

Next: Lecture 14. Synthesis & Astrobiology

Lecture notes: formingworlds.github.io/IntroductionToPlanetaryScience